

claude-windows / 45ea3a91-654d-4972-9cd3-65f35db54caf

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\45ea3a91-654d-4972-9cd3-65f35db54caf\subagents\workflows\wf_2a64bc88-2a8\agent-afbdd75859668cc68.jsonl`  
- SHA-256: `026372d598ad3bb14f115c1d03442bbd6d854ea8af97ae71fa6306cb3c15ddf1`  
- Source modified: `2026-06-10T06:10:30+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_2a64bc88-2a8`  
- session_id: `45ea3a91-654d-4972-9cd3-65f35db54caf`
```

Transcript

1. user (2026-06-10T06:09:07.932Z)

CONTEXT ? two sibling repos on a Windows machine:

- INDEXER: C:/Users/avidu/Projects/badminton-highlight-indexer ? local AI badminton(->racket-sports) highlight indexer + rally detector. FastAPI + SQLite + ffmpeg + GPU CV via WSL; Gemini = paid cloud AI. docs/ tree is rich. Current branch docs/next-steps-multisport (master = main branch).

- COLLECTOR: C:/Users/avidu/Projects/sports-data-collector ? golden-label acquisition/curation/ingestion CLI (system-of-record candidate for training corpus).

Project state (June 2026): scaffolding complete; owned-model roadmap agreed (docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md) but NO platform/owned-model implementation started; next committed platform slice = async job model. Licensing guardrail: MIT/Apache-2.0/BSD only for code+data+weights; paid LLMs are inference-only, never training-data sources. ENVIRONMENT RULES: this dev environment has NO GPU/torch/cv2 ? do NOT attempt to run the video pipeline or import torch/cv2 modules. Test suites are offline/fully-mocked and SAFE to run. You are READ-ONLY: never edit/create/delete repo files; running tests and git read commands is allowed.

QUALITY BAR: cite file paths (and line numbers where possible) for every finding. Report what IS in the code/docs, not what you assume. Be skeptical and concrete. Your final structured output is consumed programmatically by an orchestrator ? return raw data, not prose for a human.

ROLE: adversarial verifier. An auditor ("docs-strategy") claimed the finding below. Try to REFUTE it against the actual files. Default to isReal=false if the evidence does not hold up, the behavior is documented/deliberate (e.g. listed in docs/CODE_MAP.md gotchas or DEFERRED_HARDENING_PLAN.md as a known deferral ? unless the auditor adds genuinely NEW consequence), or the severity is materially overstated.

FINDING: {"title":"Licensing policy contradiction: 'Apache-2.0 only' (CLAUDE.md/PLATFORM/COMMERCIALIZATION/checkpoint) vs ratified 'MIT/Apache-2.0/BSD' (plan/NEXT_STEPS)","severity":"high","area":"licensing guardrails","file":"C:/Users/avidu/Projects/badminton-highlight-indexer/CLAUDE.md:33","evidence":"CLAUDE.md:33 ('Committed guardrails (non-negotiable)'): 'Training teachers/bases = **open Apache-2.0 only**.' Same wording in docs/archives/checkpoints/2026-06-strategy-foundation/README.md:24 ('Teachers/bases = open Apache-2.0 only'), docs/PLATFORM_ARCHITECTURE.md:269 ('Training teachers are open, Apache-2.0 models only (Qwen-VL / InternVL / Gemma 4)') and \$5A heading ~line 301 ('Base model & training (Apache-2.0 only ? licensing is decisive)'), and docs/COMMERCIALIZATION.md C14 (line 43): 'Lock to Apache-2.0; open-model teachers only'. Versus docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md:24-25: 'Commercial-clean licensing for EVERY dependency ? code, data, AND model weights: MIT / Apache-2.0 / BSD only. (Ratifies the prior \"Apache-2.0 only\" to include MIT/BSD.)' with decision row #3 'owner: yes ?' (line 152), and docs/NEXT_STEPS.md:13 'Everything stays MIT/Apache-2.0/BSD (code + data + weights)'. The archived study itself flagged the conflict (OWNED_MODEL_TRAINING_STUDY.md:132: 'guardrail literally says \"Apache-2.0 only\" but the approved list includes MIT ? ratify'). PLATFORM is also internally inconsistent: it lists InternVL (MIT) under 'Apache-2.0 models only'.\", \"why_it_matters\":\"A reader following CLAUDE.md's 'non-negotiable' guardrail would reject the MIT components the agreed plan is built on ? ActionFormer/ASFormer (MIT, the M0.4 Gen-0 substrate), InternVL3 (MIT fallback base), WASB code

(MIT). The owner-ratified policy (#61) was never back-propagated to CLAUDE.md, PLATFORM_ARCHITECTURE.md, or COMMERCIALIZATION.md C14.}"}

Steps: open the cited file(s) and any counter-evidence files; verify quotes/line claims; judge severity honestly. Cheap greps allowed; do NOT run the pipeline. Return isReal, confidence, and a 1-3 sentence verdict (with corrected detail if partially right).

2. user (2026-06-10T06:09:14.142Z)

```
1      # CLAUDE.md ? project memory / agent handoff
2
3      Local, AI-powered badminton (? racket-sports) highlight indexer + rally detector**:
4      FastAPI + SQLite + ffmpeg + GPU CV/AI, with paid Gemini as the cloud AI. Sibling
5      repos: `sports-data-collector`, a VLC-based `rally-annotator`.
6
7      ## Read these first
8      - docs/README.md ? doc index. docs/CODE_MAP.md ? code navigation + data
9        contracts (the cold-start map; read before touching code).
10     - Strategy (post-scaffolding): docs/COMMERCIALIZATION.md,
11       docs/PMF_MARKET_RESEARCH.md,
12       docs/PLATFORM_ARCHITECTURE.md, docs/DEFERRED_HARDENING_PLAN.md. Also
13       docs/ROADMAP.md
14       (offline-segmenter narrative) and docs/BACKLOG.md.
15     - History: docs/archives/ ? ADRs (decisions/DECISIONS.md), research, and
16       checkpoints/ (latest: 2026-06-strategy-foundation/). Archive policy: only move
17       terminal-state (DONE/REJECTED) work; live docs stay at docs/ top level.
18
19     ## Current state (June 2026)
20     - Scaffolding complete; the post-scaffolding strategy foundation is laid and
21       merged
22       (#52/#53/#55/#57/#58). No platform/owned-model implementation has started.
23     - Next slice = async job model (DEFERRED_HARDENING_PLAN.md #16), single-tenant.
24       Do not start implementation without explicit owner approval.
25
26     ## Architecture in one breath
27     - Pluggable registries (ABC + Registry singleton): segmenters / providers /
28       validators
29       / sports / annotations ? see backend/pipeline/segmenters/base.py. Future:
30       StorageBackend + ComputeBackend registries (PLATFORM §3).
31     - video_id = MD5 of the file ? backend/utils/hashing.py::compute_video_id (one
32       helper).
33     - Typed config = backend/config/ (Pydantic). DB schema evolves via the PRAGMA
34       user_version migration ladder in backend/infrastructure/database.py.
35
36     ## Committed guardrails (non-negotiable)
37     - Paid LLMs (Gemini/OpenAI/Anthropic) = inference / serving / fallback ONLY ? never a
38       training data source (terms/copyright). Training teachers/bases = open Apache-2.0
39       only.
40     - Owned-model base: Apache-2.0, video-capable (Qwen-VL / Gemma 4); reject Gemma
41       3
42       (viral license). Exclude minors from the training pool; per-user training data
43       needs a
44       separate explicit opt-in.
45     - v1 = rally + highlight + history + correction loop; gait/biometrics strictly
46       future
47       (GDPR Art. 9).
48     - Generic data schema; coach ? athlete is an optional overlay, not baked in.
49
50     ## Workflow conventions
51     - Branch off master; ONE PR per work item; NO PR stacking. Don't open a PR unless
52       asked.
53     - Tests: python run_all_tests.py (offline, fully mocked). Python 3.8+; ffmpeg
54       required.
55     - Never push to the default branch directly; descriptive commit messages.
56     - ?? This dev container lacks GPU / torch / cv2 / numpy ? the full pipeline can't run
57       here.
```

46 Verify pure-logic paths and hand machine-side checks (real video, GPU/WSL suite) to
the owner.
47

3. user (2026-06-10T06:09:15.107Z)

```
1      # Owned-model implementation plan (roadmap) ? DRAFT for owner review
2
3      **Status:** PLAN / roadmap. No owned-model *implementation* starts until the owner
greenlights a specific
4      milestone. Paid Gemini stays the always-ready inference/fallback throughout (never a
training source).
5      **Grounded in:** `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md`, the n=6 LOVO +
learned prototype
6      (`docs/archives/quality-iterations/2026-06-multivideo-lovo/`), and a verified strategy
workflow (2026-06-09).
7
8      ## Design principle (owner directive): *envision now, build later*
9      Every deferred piece (conversational-QA, on-device, multisport, the VLM) must be
**architecturally envisioned
10     now** so we never hit a mammoth refactor. We **design the seams** (data contracts,
export boundaries, the
11     event store) up front and **implement behind them** when each milestone is greenlit.
Punting a *feature* is fine;
12     punting its *seam* is not.
13
14     ## North Star (reframed per PR #61 review)
15     - **Immediate deliverable:** a **training framework + harness** that produces a model
with **very high precision/
16     recall for extracting highlights + rallies from a clip**. *Quality is the first-order
objective.*
17     - **Eventual product:** a **single, sport-agnostic, conversational** video model ?
upload a clip, then ask
18     contextual questions ("longest rally", "make a clip of all service faults", "rallies
over 20 shots").
19     - **On-device is LATER** (a model-compression goal so power users save upload
cost/hassle) ? definitely on the
20     roadmap, but **not** the immediate target. Don't over-index the strategy on it now.
21     - **The moat = the consented dataset + the eval harness, not the weights.**
22
23     ## Hard guardrails (non-negotiable ? "never corrupted")
24     - **Commercial-clean licensing for EVERY dependency ? code, data, AND model weights: MIT
/ Apache-2.0 / BSD only.**
25     (Ratifies the prior "Apache-2.0 only" to include MIT/BSD.) Reject viral/NC/custom-
restrictive: Gemma 3 (viral),
26     **Gemma 3n** (Gemma Terms follow derivatives ? even cleaner reject), Llama 4 (700M-MAU
cap), Commons-Clause repos.
27     - **Never train on paid-LLM (Gemini/OpenAI/Anthropic) outputs.** Also forbidden: public
grounding-CoT datasets that
28     were Gemini-generated (e.g. TVG-R1's 56K) ? we must originate our own CoT supervision
(a real, binding cost).
29     - **Exclude minors; per-user training data needs a separate explicit opt-in; split data
by match.**
30     - ? **Base-model pretraining provenance is unauditable for *every* open VLM (incl.
Qwen3-VL).** Our "no paid-LLM
31     training" rule is enforceable at *our fine-tuning data* layer, not the base's
pretraining ? an **explicit owner
32     risk-acceptance** is required to use any open VLM. (Governed by the base's license,
not our pipeline.)
33
34     ## The model strategy (verified 2026-06-09)
35     **Framework ? base** (the Gemma-4-vs-ms-swift clarification): a *framework* is the forge
(runs SFT/GRPO, emits
36     weights); a *base* is the metal you forge. You pick one framework and feed it any clean
base.
```

```

37     - **Framework: ms-swift (Apache-2.0)** ? the one forge that does native **video + audio
+ timestamp-grounding +
38     GRPO/RFT with a custom video reward** in one stack. Keep **unsloth** (Apache-2.0 core;
*avoid its AGPL Studio UI*)
39     as the 8 GB-local SFT/compression tool for the deferred on-device goal. ? Point-in-
time snapshot ? re-verify at impl.
40     - **Base (when we reach the VLM): Qwen3-VL (Apache-2.0, uniform across sizes)** ? long-
video context, purpose-built
41     **timestamp emission** (second-level localization), multi-turn video chat. Re-confirm
the exact size's LICENSE
42     pre-release. **Gemma-4 stays AMBER and bench-only** ? its Apache grant is likely but
its Prohibited-Use posture is
43     unconfirmed (counsel needed), *and* its video/timestamp capability is disputed; not
the primary extractor.
44     **InternVL3 (MIT/Apache)** = clean fallback if a Qwen blocker appears. ? base swap is
cheap *only* within
45     timestamp-capable Apache bases* (timestamp data-format + grounding-CoT are base-
specific).
46     - **START WITH THE TAL HEAD, NOT THE VLM** (the decisive call): fine-tuned VLMs **miss
strict temporal boundaries**
47     (best RL-post-trained video VLM ? R@0.7 50% on open-domain benchmarks; "coarse
regions, not accurate boundaries").
48     **Boundary accuracy IS the product.** The head (`rally_seq_proto.py`) is already de-
risked on our corpus (held-out
49     AUC 0.83?0.92), trains in minutes at ~4.5 GB on the 2070 Super ($0), is ~1M params
(trivially compressible later),
50     and **doubles as a pseudo-label teacher + the honest baseline the VLM must beat.** So:
**head now ? VLM later**,
51     strictly sequenced. When the VLM comes, prefer **two-stage** (VLM coarse-proposes,
head refines boundaries) over
52     wholesale replacement. *(Open: the VLM-ceiling number is open-domain; our amateur
footage may differ ? re-measure.)*
53
54     ## Conversational video-QA ? build the bookkeeping NOW, punt the model
55     **Punt the conversational *feature*** until extraction quality is solid (current LOVO
~0.583 ? "very high"; a chat UI
56     over a weak extractor exposes the weakness). **Reject end-to-end-VLM-answers-each-turn**
(expensive; bad at exact
57     counting; the strong teachers are license-forbidden). **But build the *seam* now** (the
brief requires it):
58     - **Architecture = "extract once, query many":** EXTRACTION (TAL head ? later VLM) emits
structured timestamped
59     spans ? a durable **per-video EVENT STORE** ? a **structured QUERY layer** (text-to-
SQL + ffmpeg clip-cut). The
60     cited queries ("longest rally", "all service faults", "rallies > 20 shots") are
**structured-numeric ? SQL, not a

```

4. user (2026-06-10T06:09:20.584Z)

```

255     **answers questions about rallies, shots, and technique**. This is the conversational
256     brain of the "assistant coach." Crucially, it does **not** replace the known stack; it
257     is introduced **behind the same abstraction, with the paid API as an always-ready
258     fallback**.
259
260     > **The moat is *not* the model.** Anyone can download Qwen-VL/Gemma. The defensible
261     > assets are ** (a) the consented, badminton-specific video?QA dataset** and ** (b) the
262     > eval harness**. Build and message *those* ? not "we own an LLM."
263
264     > **? Committed guardrail (legal, non-negotiable).** We respect the licensing findings
in
265     > full. **Paid frontier models (Gemini / OpenAI / Anthropic) are used for *inference /
266     > serving / fallback* only ? never as a *training* data source.** Harvesting their
outputs
267     > to train our owned model would breach their "no competing model" terms (a
terms/copyright

```

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268     > violation), so it is prohibited in both engineering and legal process. **Training
teachers
269     > are open, Apache-2.0 models only** (Qwen-VL / InternVL / Gemma 4). Detail in the
270     > distillation constraint below.
271
272     ### Core principle: AI is a pluggable provider with a paid fallback
273     The codebase already abstracts the cloud AI behind `provider_registry` /
274     `AIVideoProvider` (`backend/providers/`). Make that a **first-class platform
principle**:
275     - Every generative capability routes through the **provider registry**.
276     - The **owned model** plugs in as a new provider (`owned_vlm`); **Gemini (paid, via
277       Vertex AI)** stays registered as the **fallback** (failover on error/low-confidence,
278       or as the quality ceiling/teacher reference).
279     - This lets us **ship today on the known paid stack** and **swap in the owned model when
280       its quality clears a measured bar** ? with zero architectural churn and the paid path
281       always one config flip away. (Mirrors the COMMERCIALIZATION C4 A/B/C strategy: the
282       owned model is the long-run "(A) eliminate the per-call dependency"; paid is "(C)".)
283
284     ### The model / training layer (new sub-layer, fed by the scaffolding)
285     ```
286     Data curation      Training      Eval      Serving
287     (consented user vid (fine-tune an open, (extend the existing (self-host inference
288     + golden set +      commercially-usable eval harness to QA as a provider /
289     rally/segment/event base VLM via      quality + the      ComputeBackend;
290     annotations ?      LoRA/QLoRA)      regression gate)      paid fallback)
291     video-QA pairs)
292     ```
293     This layer **reuses every other layer**: **Storage** holds the training corpus;
294     **Data** supplies labels (the `annotation` registry + `candidate_segments` +
295     golden set are already the labeling seam); **Compute/Orchestration** runs training and
296     inference jobs (even **BYO-compute** can serve inference on the tenant's GPU); the
297     **eval harness** already in `backend/eval/` extends to grade QA quality. **In other
298     words, the rock-solid scaffolding is exactly what makes the model flywheel possible** ?
299     which is why scaffolding comes first (see §6).
300
301     ### Base model & training (Apache-2.0 only ? licensing is decisive)
302     **Recommended near-term base: `Qwen2.5-VL-7B-Instruct`** ? Apache-2.0, native long-video
303     (dynamic-FPS sampling, temporal mRoPE), mature LoRA tooling. Upgrade path
304     **Qwen3-VL-4B/8B (Apache-2.0)**; **Gemma 4 multimodal (Apache-2.0, 2026)** is a clean
305     Google-stack alternative to evaluate.
306     - ?? **Lock to Apache-2.0-clean bases** so you can fine-tune, **keep weights private,
307       gate behind subscriptions, and never owe open-sourcing or branding**. (Note Qwen2.5-VL
308       is Apache **only at 7B/32B** ? 3B/72B use the custom Qwen license.)
309     - ? **Reject for the flagship:** **Gemma 3** ? custom terms with a **viral derivative
310       clause** (anything fine-tuned on it, *or even trained on its synthetic outputs*, stays
311       permanently bound to Google's *updatable* Prohibited-Use Policy); **Gemma 4's switch
312       to
313       Apache-2.0 removes this**. **Llama 3.2 Vision** ? custom license forces a "Llama-?"
314       model name + "Built with Llama" badge and a 700M-MAU ceiling. **`gpt-oss`** ?
Apache-2.0
314     but **text-only**, so usable only as an optional text-reasoning layer, never the video

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5. user (2026-06-10T06:09:21.993Z)

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35     | C5 | **Output codec royalties** | **P0** | ? | Highlights are still encoded **H.264 +
AAC** (`libx264` in `backend/pipeline/stitcher/compiler.py`) ? encode-side AVC patent exposure
remains. | Switch output to **AV1 + Opus** (royalty-free) on an **Ada-class GPU (L4/L40)** for
HW encode. See §4. Decision: provision Ada-class GPU? |
36     | C9 | **Packaging / distribution** | **P0** | ? | No `pyproject.toml`; the diagnostic
`setup.py` shadows the build system. | Design in `DEFERRED_HARDENING_PLAN.md` (#7) ? **awaiting
your approval** before code. |
37     | C12 | **SaaS foundations (async/auth/tenancy)** | **P1** | ? | Sync `/api/process`
blocks on long videos; no auth/tenancy. | Multi-user/platform strategy in
`PLATFORM_ARCHITECTURE.md` (pluggable storage/compute, tenancy, orchestration); async slice =
`DEFERRED_HARDENING_PLAN.md` (#16). |

```

38 | | **? IN-FLIGHT (in progress / watch)** | | | |
39 | C4 | **Runtime AI strategy (A/B/C + owned model)** | **P0** | ? | Live = **(C) Gemini 2.5 Flash, paid tier**. Long-term = **own a self-hosted generative video-QA model** (fine-tuned open VLM) as primary, **paid Gemini always-ready as fallback** via the provider registry. | Make "provider registry + paid fallback" a core principle; build the owned model **after** the scaffolding/flywheel are solid. See `PLATFORM\_ARCHITECTURE.md` §5A. |
40 | C13 | **Beachhead validation** | **P0** | ? | Market research (`PMF\_MARKET\_RESEARCH.md`) recommends **India badminton academies** as the #1 beachhead. Unvalidated. | Cheapest tests: 10?15 academy/coach discovery interviews; rally-detection P/R benchmark on 100+ real amateur clips; fake-door pricing (B2C ~\$99/yr, B2B academy ~\$600/yr). See §6. |
41 | C8 | **Golden-set vendoring spend** | **P1** | ? | ADR-006: **Roora (Bengaluru)** selected for an initial paid pilot; licensed/owned video only (no user uploads to vendors). | Confirm quality-per-dollar from the pilot; then GS-3 `HumanVendorProvider`. See `GOLDEN\_SET\_VENDORING.md`. |
42 | C3 | **Shuttle-tracking weights provenance** | **P1** | ? ? | WASB-SBDT **code is MIT (clear)**; distributed **checkpoints are academic-data-trained** ? not covered by the MIT code grant. MonoTrack/YOLO-NAS banned (noncommercial). | Train-own weights (Phase-4 roadmap) **or** confirm per-checkpoint terms ?? before commercial deploy. |
43 | C14 | **Owned-model base licensing + training-data policy** | **P1** | ? ? | Owned video-QA model (PLATFORM §5A) must build on **Apache-2.0-clean, video-capable** bases ? **Qwen2.5-VL-7B/32B** (Apache; 3B/72B are *not*), **Qwen3-VL-4B/8B**, or **Gemma 4** ? to keep weights private, gate, and monetize. **Reject Gemma 3** (viral derivative clause ? permanent Google PUP), Llama Vision (naming + 700M-MAU), `gpt-oss` (text-only). **Hard rule: never train on paid-LLM (Gemini/OpenAI/Anthropic) outputs** (competing-model clause). The moat is the **consented dataset + eval harness**, not the model. | Lock to Apache-2.0; **open-model teachers only**; separate explicit **training opt-in**; **exclude minors** from the training pool; per-clip consent ledger. See PLATFORM §5A/§7. |
44 | C6 | **Inbound HEVC decode** | **P2** | ? ? | Decoding users' GoPro H.264/**HEVC** uploads is unavoidable; HEVC pools are fragmented + highest-uncertainty. | Low/uncertain residual; confirm decode-side exposure with counsel before launch. |
45 | C7 | **Hosting unit-economics** | **P2** | ? | Per-video cost is *cheap* for Gemini (~\$0.05 input for a 10-min clip); risk is volume × latency, not unit price. Self-host/offline costs unmeasured. | **Benchmark \$/video & latency on the target VM** for A/B/C before pricing. See §5. |
46 | C10 | **Value layer (product surface)** | **P2** | ? | Minimal exports shipped (EDL/JSON/CSV + `basic\_stats`, PR #51). | Deepen: per-rally tags/notes that feed golden, per-player/match stats, clip gallery. See §6. |
47 | C11 | **Input-quality enforcement** | **P2** | ? | Per-video observability reports surface a customer verdict + footgun flags incl. audio-present (PR #50). | Turn more `VIDEO\_RECORDING\_GUIDELINES` dimensions into validators/report rules. |
48 | C2 | **supervision` ByteTrack pin\*\* | \*\*P2\*\* | ? | Pinned `<0.30.0` (deprecation ahead). | Migrate before lifting the pin (BACKLOG). Not a legal risk, a maintenance one. |
49 | | **? COMPLETED** | | | |

6. user (2026-06-10T06:09:22.297Z)

1 # Next steps (live) ? owned-model + multisport
2
3 **Updated:** 2026-06-09 (end of the quality-iteration + multisport-recon session).
Authoritative roadmap:
4 `docs/OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md`. **Latest blow-by-blow:** memory `current-state.md` / `session-handoff.md`.
5 This file = the prioritized "what to do next," refreshed each session.
6
7 ## The recommendation (one paragraph)
8 Build the **quality-first extraction harness** before anything else: a learned **TAL head** (ActionFormer/ASFormer, MIT)
9 over trajectory features is the robust path ? it **decouples from the multi-court "background games" confound** that the
10 unsupervised heuristic (and audio, and spatial court-gating) can't escape (all three tested + failed; it's a *temporal*
11 rally-vs-non-rally problem). **Framework = ms-swift, base = Qwen3-VL** (Gemma-4 AMBER/bench) come **later** for the
12 conversational/grounding upgrade ? start with the head. **On-device is deferred**

```

(export-clean ops now so it stays
13     reachable). Everything stays **MIT/Apache-2.0/BSD** (code + data + weights). Bottleneck
= **data + per-sport trajectory
14     detectors**, not model architecture.
15
16     ## Latest findings (2026-06-09)
17     - **6-match badminton LOVO:** learned prototype **0.583** vs heuristic **0.480**
(learned leads since n=3, AUC 0.83?0.92,
18     decoupled from multi-court). The "contrast" metric (in-rally÷dead-time visibility)
predicts heuristic F1 monotonically.
19     - **?? Multisport (Roora pickleball + TT):** labels **ingest cleanly** (`curate convert-
cvat` ? 6 rallies/sport, sane).
20     **WASB-transfer SURPRISE (directional, n=6 each ? TINY):** the badminton shuttle
detector segments **TABLE TENNIS at
21     F1 0.909** out-of-the-box (!) but **pickleball only 0.308**. ? **The per-sport-
detector blocker is SPORT-DEPENDENT:** TT may
22     be servable with WASB short-term; pickleball needs a proper (MIT/Apache) ball
detector. Confirm/deny with more videos.
23     - **Owner field notes (2026-06-09):** camera height/distance **varied on purpose**
(great generalization data); **multi-court
24     adjacent-court SOUND dominates the signal** ? empirically reconfirms audio-foreground
isolation won't work + the
25     robustness-first learned-model path. Multi-court is the realistic (academy)
environment, so this *is* the target distribution.
26
27     ## Prioritized next steps
28     1. **[owner greenlight] Phase-0 (M0.x), $0/local:**
29     - **M0.4 ? Gen-0 TAL harness** (first-class): productionize `rally_seq_proto.py` ?
one-command train?eval?**ship-gate** over
30     ActionFormer/ASFormer, reusing `metrics.match_intervals`. **Define the ship-gate
target** (explicit F1@10U; floor = beat

```

7. user (2026-06-10T06:09:31.972Z)

```

15     - **COMMERCIALIZATION.md** ? commercial tracker (dashboard C1?C14, grouped Pending ? In-
flight ? Completed with a P0/P1/P2 column).
16     - **PMF_MARKET_RESEARCH.md** ? cited market study; beachhead = **India badminton
academies**.
17     - **PLATFORM_ARCHITECTURE.md** ? multi-user "assistant coach" platform: pluggable
`StorageBackend`/`ComputeBackend` (BYO), generic schema (coach overlay optional), owned video-QA
model layer ($5A), training-backend survey + nanochat-style harness, committed legal guardrail,
per-user-adapter direction, v1-vs-vision scoping.
18     - **DEFERRED_HARDENING_PLAN.md** ? packaging, train/eval guardrails, async job model
(#16) ? *pending approval*.
19     - **CODE_MAP.md / BACKLOG.md / ROADMAP.md** ? updated for the config/identity cleanup.
20
21     See git history around #52/#53/#55/#57/#58 for exact snapshots.
22
23     ## Committed decisions / non-negotiables recorded here
24     - **Paid frontier models (Gemini/OpenAI/Anthropic): inference / serving / fallback only
? never a training data source** (competing-model terms). Teachers/bases = open Apache-2.0 only.
25     - **Owned-model base = Apache-2.0-clean, video-capable** (Qwen2.5-VL-7B/32B, Qwen3-VL,
Gemma 4); **reject Gemma 3** (viral license clause); **minors excluded** from the training pool;
per-user training needs a separate opt-in.
26     - **v1 = rally detection + highlight compilation + per-user history + coach/user
correction loop** ("assistant-coach lite"); **gait/biomechanics strictly future** (Art. 9).
27     - **Generic data schema** (User ? VideoCollection ? Video ? Highlight ?
HighlightMetadata); coach ? athlete is an **optional overlay** (a market-fit hypothesis).
28
29     ## What comes next (per owner)
30     - First implementation slice = **async job model (DEFERRED #16)**, single-tenant ?
**pending explicit go-ahead** (no implementation started without approval).
31     - Quality iterations with new labeled videos; golden-set expansion + the **golden QA
eval set** (the prerequisite for the owned-model training harness).
32

```

33 When the first implementation slice or a quality milestone completes, add a new dated
34 checkpoint here (or under `quality-iterations/`).

8. user (2026-06-10T06:09:32.919Z)

125 model trained on provenance-unclear features **does not launder the provenance.**
ROADMAP Phase-4 "train own WASB
126 weights" is a **prerequisite, not an aside.**
127 - **Gemma 4 "clean + native-video" rests on post-cutoff secondary sources** ? keep AMBER
until a raw HF LICENSE
128 revision is verified, given the Gemma-3 rejection. Its timestamp-emission recipe is
less documented than Qwen's.
129 - **Existing distilled model was trained on Gemini silver labels** ? the (correct) eval-
only policy now **forbids
130 this; that pipeline must be purged of Gemini-derived training labels** to comply.
131 - **Policy ratifications needed:** guardrail literally says "Apache-2.0 only" but the
approved list includes MIT ?
132 ratify "Apache-2.0/MIT". The MAU-cap rejections (72B/78B) may be over-conservative at
our scale.
133 - **On-device privacy claim is unbenchmarked** ? no tokens/sec for a quantized 4B VLM on
8 GB Turing at
134 1080p/60fps. **Measure before relying on it.**
135 - **Amplification is load-bearing for Gen-2** ? 128 raw rallies can't reach the ~5k?20k
clip?QA band alone; if
136 augmentation + QA-multiplexing + pseudo-labels underdeliver, Option B is under-fed.
Amateur 60fps may need more
137 labels than broadcast-trained literature implies.
138 - **Dropped option worth keeping:** InternVideo2 encoder (MIT) + light grounding head ?
the *only* data-realistic
139 owned-video experiment available at n?128 today. And: a **one-time ~\$1?2k 24 GB local
card** (e.g. used 3090)

9. user (2026-06-10T06:09:33.918Z)

140 our own CoT supervision). **Two-stage** (VLM proposes, head refines). Cloud train on
Modal (no-access) / RunPod-Secure.
141 - **Hard pre-serving gates (unmeasured, load-bearing):** the **\$/call crossover** (self-
hosted VLM vs Gemini) **a real
142 VRAM/token + cost benchmark** on rented A100/L40S (ms-swift multimodal-GRPO examples
use 6?8 GPUs ? **budget is the
143 largest open risk and is currently unquantified**). Don't stand up an owned serving
stack until these clear.
144
145 ---
146
147 ## Decisions / status
148 | # | item | status |
149 |---|---|---|
150 | 1 | Greenlight Phase-0 milestone(s) ? rec: **M0.1 corpus+event-index**, then **M0.4
harness** | awaiting owner |
151 | 2 | WASB C3 ? first-class action item (own-weights / clean swap) | **owner: yes** ? |
152 | 3 | Ratify licensing **MIT / Apache-2.0 / BSD** (code+data+weights) | **owner: yes** ?
|
153 | 4 | Collector = corpus + event-store system-of-record | **owner: confirm?** |
154 | 5 | Rename repo off "badminton" | **owner action** ? |
155 | 6 | Conversational-QA: build seam now, punt model | **rec accepted?** |
156
157 ## Open risks / unknowns (none block Phase-0 research)
158 Budget/cost to reach Gen-2 **unquantified** (largest risk) · ship-gate "very high P/R"
target undefined · per-sport clean
159 trajectory detectors not yet identified · base-model pretraining provenance unaudit-
able (risk-acceptance) · real VLM
160 VRAM/token cost · single-model-multisport-temporal generalization unproven · DPA + no-
train (minors-excluded) for cloud
161 training · on-device 4B latency on 8 GB unmeasured.

162

163 ---

164 *Design-only and reversible. Nothing starts until you greenlight a milestone. Sequencing discipline holds: the committed

10. user (2026-06-10T06:10:23.453Z)

Structured output provided successfully

11. assistant (2026-06-10T06:10:30.758Z)

Verification complete. The finding is real: all cited quotes hold at the cited locations, the contradiction between the live "non-negotiable" docs (CLAUDE.md, PLATFORM_ARCHITECTURE.md, COMMERCIALIZATION.md C14) and the owner-ratified MIT/Apache-2.0/BSD policy is genuine, with the minor corrections that the archived checkpoint is a frozen snapshot and the plan itself documents the ratification ? making this a known-but-unpropagated policy update rather than an undocumented conflict.